

Hein Zarni Aung

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EDUCATION

Stony Brook University

Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Systems Fundamentals I & II, Algorithms, Programming Abstractions

RESEARCH

Undergraduate Research Assistant — Ethos Lab, Stony Brook University

Sep 2025 - Dec 2025

- Built a real-time pipeline to monitor newly issued SSL/TLS certificates and flag fraudulent domains impersonating EZ-Pass tolling services, using Selenium for automated scraping and Python-based heuristics to triage flagged sites.

INTERNSHIPS

Software Engineering Intern — Clustr

Jan 2026 - Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows and activity feed endpoints.
- Implemented secure authentication features, including a password-reset flow (client screen plus backend cloud function) and custom claims to drive new-user onboarding logic.

Software Engineering Intern — Prenora Dynamics LLC

Jun 2026 - Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for a real-time, low-latency video chat feature built in-house, researching and evaluating SDK approaches to replace third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, and built a seed-data workflow to streamline debugging and local development.

CAMPUS INVOLVEMENT

Teaching Assistant — Data Structures & Algorithms, Stony Brook University

Aug 2025 - Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

PROJECTS

Discord Voice Channel Activity Monitor (Node.js, Discord.js, PM2, AWS)

- Built a real-time event-driven Node.js/Discord.js system that detects voice-channel state changes and pushes instant notifications to users across a distributed friend group.
- Implemented customizable DM notifications, VIP lists, and per-server/per-user channel thresholds with JSON-based configuration.
- Deployed on a cloud server using PM2; reached 50 concurrent active users.

SKILLS

Languages: Java, Python, C, JavaScript, TypeScript

Backend: Node.js, Express.js, FastAPI

Databases & Cloud: PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP

Tools: Git, Docker, Linux, PM2, Selenium